

Maha Trabelsi

Master's Student in Data Science — AI for Development & Public Impact

✉ maha.21.trabelsi@gmail.com ☎ +216 21 535 271 in LinkedIn 🐙 GitHub

Profile

Master's student in Data Science specializing in Artificial Intelligence, with a strong interest in applying data-driven solutions to real-world development challenges. Experienced in building AI systems for healthcare, decision support, and resource optimization, with a focus on impact-driven applications in public health, sustainability, and underserved communities. Motivated to contribute to international development initiatives through data analysis, innovation, and interdisciplinary collaboration.

Education

Higher Institute of Computer Science Ariana (ISI Ariana) <i>Master's Degree (Research) in Data Science</i> Focus: Artificial Intelligence, Machine Learning, Data Analysis	<i>Ariana, Tunisia</i>
Higher Institute of Computer Science Ariana (ISI Ariana) <i>Bachelor's Degree in Computer Science</i> Specialization: Software Engineering and Information Systems	<i>Ariana, Tunisia</i>
Taha Hussein Megrine High School <i>Baccalaureate in Informatics</i>	<i>Megrine, Tunisia</i>

Selected Projects

AI-Based Medical Triage System ◦ Designed an AI-assisted decision-support system for low-resource healthcare settings, enabling structured patient triage based on reported symptoms. ◦ Applied NLP and rule-based reasoning to improve accessibility and decision-making in underserved environments. ◦ Focused on scalability, interpretability, and usability in rural and resource-constrained contexts.	GitHub 
Critical Power Intelligence System ◦ Developed a decision-support system to simulate resource allocation in critical infrastructure scenarios. ◦ Modeled optimization strategies for energy distribution under constraints, reflecting challenges in resource-limited environments. ◦ Demonstrated the use of AI-driven approaches for operational planning, resilience, and emergency response. ◦ Built a full-stack application using FastAPI and React, integrating backend logic with interactive interfaces.	GitHub 
DiaGraph – Node Classification on Noisy Graphs ◦ Developed a graph machine learning framework to study node classification under noisy and imbalanced conditions. ◦ Evaluated robustness using Macro F1-score, emphasizing performance across minority classes. ◦ Implemented an automated benchmarking and leaderboard system using GitHub Actions. ◦ Strengthened skills in reproducible evaluation, collaborative development, and model assessment.	GitHub 
SMS Spam Classifier ◦ Designed an NLP pipeline using TF-IDF vectorization and Logistic Regression for text classification. ◦ Performed data preprocessing, feature extraction, and model evaluation in Python.	GitHub 

Experience

Mobile and AR Development PFE Intern <i>Octomiro</i> ◦ Contributed to the development of a cross-platform mobile application using Flutter and Unity. ◦ Integrated AR features and supported real-time data management with Firebase services. ◦ Collaborated in a technical team environment, improving software delivery and problem-solving skills.	<i>Tunis, Tunisia</i> <i>Feb 2025 – Jun 2025</i>
Frontend Developer Intern <i>SASCODE</i>	<i>Lac, Tunisia</i> <i>Mar 2024 – Jun 2024</i>

- Developed user-facing web interfaces with React.js for booking and reservation platforms.
- Integrated backend APIs and contributed to responsive, functional web experiences.
- Improved ability to translate user needs into accessible and modern digital interfaces.

Frontend Developer Intern

SASCODE

Tunis, Tunisia
Jul 2024 – Aug 2024

- Developed the frontend interface for the TPC Tennis Tournament platform using React.js.
- Built responsive pages and interactive UI components for tournament and event management.
- Collaborated with the backend team to integrate APIs and ensure smooth platform functionality.

Leadership and Volunteering

- **International Relations Leader**, Tunisian Scouts (2018–2025)
Coordinated international collaborations, exchanges, and communication with international scouting organizations and partners.
- **Safe from Harm Team Leader**, Tunisian Scouts (2018–2025)
Led initiatives focused on youth protection, safety awareness, and the implementation of safeguarding practices during scouting activities and events.
- **Scout Leadership and Community Engagement**, Tunisian Scouts (2018–2025)
Contributed to leadership development, event coordination, and community-based initiatives promoting teamwork, social responsibility, and youth empowerment.
- **Volunteer**, Tunisian Red Crescent (2018–2025)
Participated in humanitarian and community support activities, gaining exposure to field-level challenges and public service environments.
- **Chairwoman**, ISI Geeks (2023–2024)
Led student activities, coordinated teams, and organized academic and technology-focused events.
- **General Secretary**, IEEE ISI Student Branch (2023–2024)
Supported administrative coordination, communication, and planning of student-led technical initiatives.

Selected Academic Programs & Certifications

- **RISE – Graph Neural Networks for Rising Stars (2026)**
Advanced research-oriented program focused on Graph Neural Networks, representation learning, and AI applications.
- **DAAD Arab-German Summer School on Biomedical Imaging with Deep Learning (BILD2025)**
International summer school focused on deep learning applications in medical imaging and AI for healthcare.
- **Azure AI Fundamentals** – Microsoft
- **Machine Learning with Python** – IBM / Coursera
- **Python for Data Visualization** – LinkedIn Learning

Skills

- **Programming:** Python, Java, JavaScript, SQL, Kotlin
- **Data and AI:** Machine Learning, Statistical Modeling, Data Analysis, NLP, scikit-learn, PyTorch, Keras, Power BI, Excel
- **Development Tools:** Git, GitHub, Firebase, MongoDB, SQLite, Postman
- **Web and Mobile:** React.js, Next.js, Flutter, Node.js, FastAPI, PHP, HTML/CSS
- **Soft Skills:** Teamwork, leadership, communication, project coordination, adaptability

Languages

- **Arabic:** Native
- **English:** Full Professional Proficiency
- **French:** Working Proficiency
- **Spanish:** Basic Knowledge